EXP.NO: 4
DATE: 30/7/26

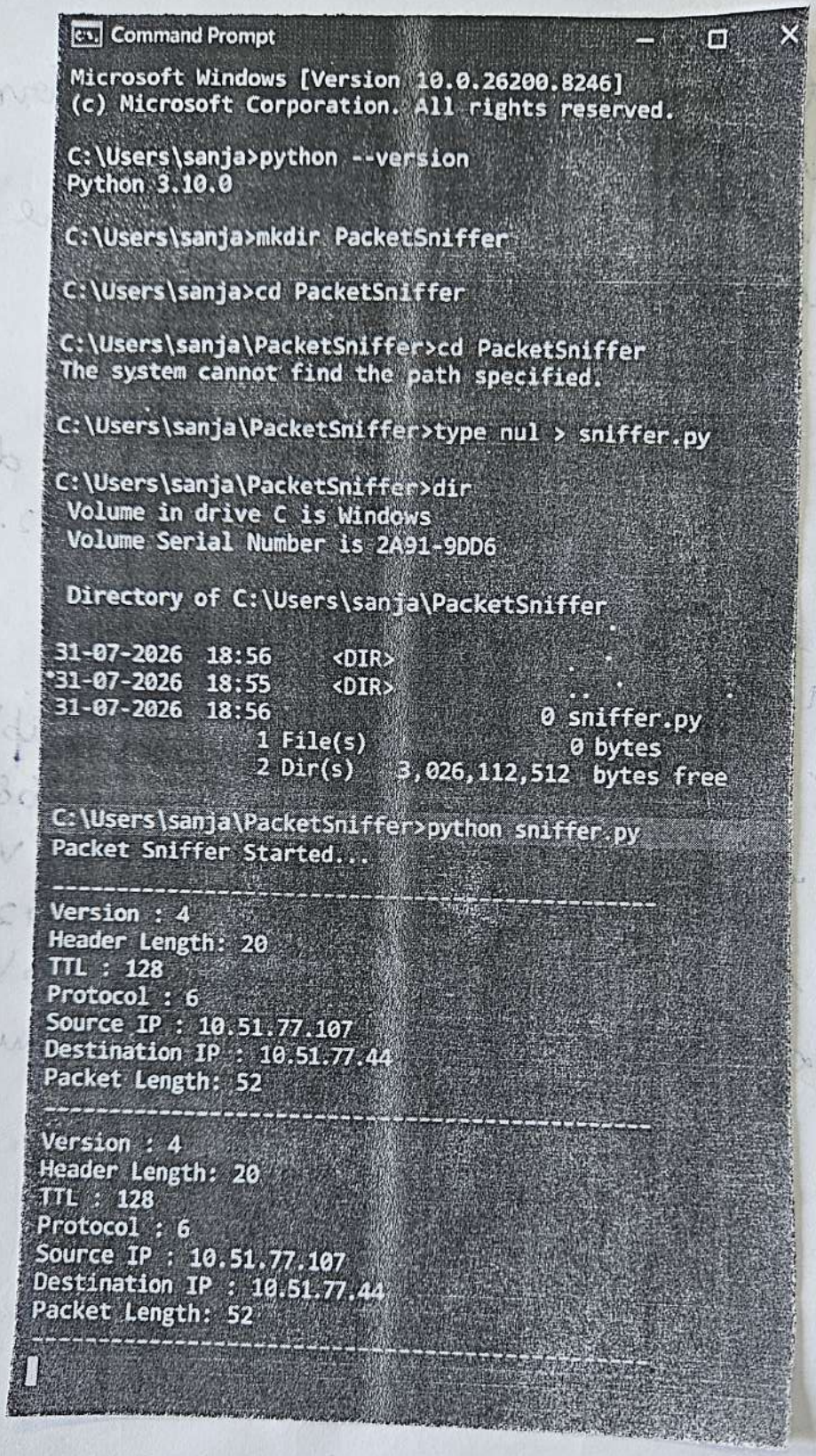
PACKET SNIFFING USING RAW SOCKETS IN PYTHON.

Aim:

To implement a packet sniffer using raw sockets in python to capture live network traffic and parse ethernet IP, and transport layer headers.

Procedure:

1. Install python 3 on a linux system or VM.
2. Open terminal with root privilages
3. Create a python file named sniper.py
4. Import required modules
5. Create a raw socket using AF_PACKET and SOCKET_RAW
6. Capture incoming ethernet features from network.
7. Parse ethernet header to obtain source MAC, destintion MAC and protocol type.
8. If protocol is IPV4, parse IP header.
9. Extract IP version, header length, TTL, protocol number, source IP.
10. Display captured packet information on the screen.


Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.
C:\Users\sanja>python --version
Python 3.10.0
C:\Users\sanja>mkdir PacketSniffer
C:\Users\sanja>cd PacketSniffer
C:\Users\sanja\PacketSniffer>cd PacketSniffer
The system cannot find the path specified.
C:\Users\sanja\PacketSniffer>type nul > sniffer.py
C:\Users\sanja\PacketSniffer>dir
Volume in drive C is Windows
Volume Serial Number is 2A91-9DD6
Directory of C:\Users\sanja\PacketSniffer
31-07-2026 18:56 <DIR> .
31-07-2026 18:55 <DIR> ..
31-07-2026 18:56 0 sniffer.py
1 File(s) 0 bytes
2 Dir(s) 3,026,112,512 bytes free
C:\Users\sanja\PacketSniffer>python sniffer.py
Packet Sniffer Started...
Version : 4
Header Length: 20
TTL : 128
Protocol : 6
Source IP : 10.51.77.107
Destination IP : 10.51.77.44
Packet Length: 52
Version : 4
Header Length: 20
TTL : 128
Protocol : 6
Source IP : 10.51.77.107
Destination IP : 10.51.77.44
Packet Length: 52

sniffer.py - Notepad

File Edit View

```
import socket
import struct

# Create raw socket
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

# local machine IP
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))

# Include IP headers
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

# Enable promiscuous mode (Windows)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

print("Packet Sniffer Started...\n")

try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)

        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)

        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9]
        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination IP :", destination)
        print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

Result:

The packet sniffer was successfully implemented using python raw sockets. It captured live network packets and correctly extracted and displayed ethernet and IPv4 header information, including MAC address, IP address, TTL and transport layer protocol details.